



ElectronicsTutorials

DC CIRCUIT THEORY

For Students, Professionals
and Beyond

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1. THE STRUCTURE OF MATTER

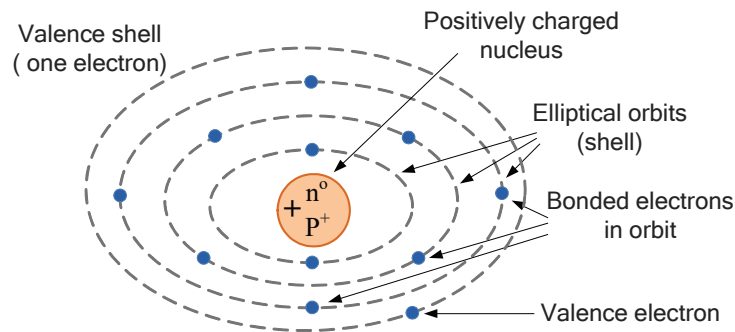
Matter is everywhere in the form of a solid, gas, or liquid, with the basic building blocks of matter being elements. Every element consists of just one distinct type of atom and it is this atom that gives each element its own particular characteristic.

Matter is not a solid structure but are composed of three fundamental particles: neutrons, protons, and electrons arranged in various combinations. It is the number of these neutrons, protons, and electrons that makes each atom different from another one.

Each atom of matter has a positively charged nucleus consisting of protons and uncharged particles called neutrons. The protons and neutrons are grouped together in the center of the atom, while the electrons are arranged in different elliptical orbits, called shells, around the nucleus.

Electrons in different orbits can rotate around the nucleus in different directions and distances from the nucleus, thus producing a three-dimensional atom similar to that shown in Figure 1.

FIGURE 1. STRUCTURE OF AN ATOM



Then matter is electrical in nature as it contains particles of electricity in the form of protons and electrons. The negatively charged electrons in the nearest orbit have a greater force of attraction to the positively charged nucleus full of protons.

Electrons in the farthest orbit have the least force of attraction and are loosely held to the nucleus are called valence electrons and therefore rotate around the outermost valence shell.

The number of protons present within an atoms nucleus specifies its atomic number. For example, the atomic number of a copper atom is 29, that is it has 29 electrons circulating its nucleus including a single valence electron in its outermost shell. It is an atoms atomic number which distinguishes it from another.

In a single atom the number of negatively charged electrons and positively charged protons are the same, making it electrically neutral.

However, if the loosely held electron in the outermost valence shell receives a stimulus from an external energy source, for example heat, or an electrical voltage, this will cause them to break free from its bond and move randomly around through the space in between the various shells of the other neighbouring atoms.

An atom's outermost shell is called its Valence Shell with electrons orbiting this shell called Valence Electrons

These loose electrons are called “free electrons” and together constitute a negative electrical charge. Therefore, the ability of an atom to either gain or lose one or more of its outermost free electrons increases if the valence shell is located farther enough away from the atoms nucleus. Materials which contain a large number of free electrons are called conductors.

Note that the basic structure of the atom shown in Figure 1. with electrons circulating its nucleus is commonly called the Bohr Model. This simple depiction is sufficient for showing and describing the properties of atoms here.

2. ELECTRICAL CONDUCTORS

While free electrons are allowed to move at random from one atom to another within a material, metallic elements have the ability to release their valence electrons more easily because they are more loosely bound to their nucleus.

Most materials made of metals are considered to be good conductors of electricity as they have few valence electrons in their outermost shell allowing the free electrons in the form of an electric current to easily flow between the adjacent atoms.

Materials made of metal have a low resistance and high conductance. Conductance (G) is equal to the reciprocal of resistance and is a measure of how good a metal conductor is at carrying an electric current.

A conductor contains mobile charges which can easily move through a material. An insulator has no mobile charges

Examples of good conductors are generally metals such as copper, aluminium, gold, silver or metal alloys, non-metals such as carbon, graphite, mercury and some conductive polymers, acids and salts.

Then good conductors such as metals contain large numbers of free electrons. Most electrical cables and cords have copper wires inside of them, since copper is a good conductor of electricity and flexible.

So if a voltage source is connected to one end of an electrical conductor, and a lamp to the other end, free electrons under the influence of the voltage source will readily flow through the conductive wire, thus constituting a flow of electric current from the negative end to the positive end of the source, and the lamp will light up.

3. ELECTRICAL INSULATORS

While electrical conductors are very good at passing an electric current, insulators are not making insulators the exact opposite of conductors. An insulator is a material with virtually no or very few loose or free electrons and therefore offers a very high resistance and low conductance to the flow of current.

This is because their outermost valence shell has a high number of electrons that are tightly bonded to the nucleus orbiting around it. So therefore, the outermost valence electrons do not overlap with neighbouring atoms making it very difficult for the electrons to move freely between them.

Examples of insulators include glass, porcelain, rubber, pvc (polyvinyl chloride) plastics, mineral oils, dry wood or paper, etc. Ceramic is one of the best insulators.

While electrical cables and cords have conductive copper wires inside of them, they are covered with insulating materials on the outside as shown in figure 2. This allows electrical cables to be safely touched, even when they are carrying electricity.

FIGURE 2. COPPER CONDUCTOR WITH INSULATION



Then insulators are very poor conductors of electricity as they have very few free electrons in their valence shell.

The basis for selecting a particular insulating material over another, generally depends on the breakdown voltage rating of the cable or circuit in which it is to be used, as well as material flexibility, environmental conditions (water, chemical exposure, etc.) and the temperature to which the insulating material will be subjected to during its life span.

Thus while we need conductors in electrical engineering to carry current to the different parts of a circuit, we also need insulators to stop those conductors from shorting together.

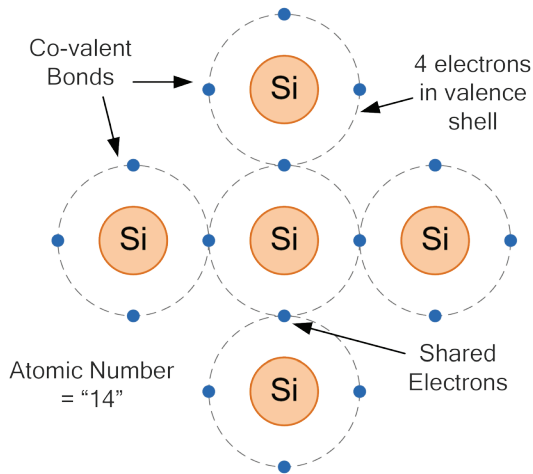
4. SEMICONDUCTORS

As well as conductors and insulators, there is a third type of material called a semiconductor. Semiconductor materials such as Silicon (Si), Germanium (Ge) and Gallium Arsenide (GaAs), have electrical properties somewhere between those of a conductor and an insulator. The result is that they are neither a good conductor, nor are they good insulators (hence their name “semi”-conductors).

Semiconductors have very few free electrons orbiting around in their valence shell because their atoms are closely grouped together in a tight crystalline pattern called a “crystal lattice”.

However, their ability to conduct electricity (conductivity) can be greatly improved by adding certain “impurities” to their crystalline structure thereby, producing more free negatively charged electrons than positive holes or vice versa.

FIGURE 3. SILICON MATERIAL



A semiconductor material such as silicon as shown in Figure 3 or germanium which is pure and contains no impurities is known as an intrinsic semiconductor as the number of free electrons and holes are equal.

So by controlling the amount of impurities added to this intrinsic semiconductor material it is possible to control its electrical conductivity to any level between that of a conductor and an insulator.

The impurities are called donors or acceptors depending on whether they

produce electrons or holes respectively. This process of adding impurity atoms to semiconductor atoms (the order of 1 impurity atom per 10 million (or more) atoms of the semiconductor) is called Doping.

Semiconductors on their own are neither good conductors nor good insulators

By doping silicon with “pentavalent” (5-electron) impurities such as Antimony (symbol Sb) or Phosphorus (symbol P), the resulting semiconductor material will have an excess of current-carrying electrons, each with a negative charge. Therefore, it

produces an “N-type” semiconductor material.

Doping silicon with “trivalent” (3-electron) impurities such as Boron (symbol B) or Indium (symbol In), the resulting semiconductor material will have a large number of holes in its structure making it more positive and therefore producing a “P-type” material.

Then by using different doping agents to a base material of either Silicon (S) or Germanium (Ge), it is possible to produce different types of basic semiconductor materials, either N-type or P-type. Electrical and electronic devices, including diodes, transistors, integrated circuits and light-emitting diodes, etc. operate over a wide range of current- and voltage-handling capabilities, and all made from silicon dioxide sand.

5. ELECTRIC CURRENT

Electrical current is the flow or movement of charged particles along conductors used within a closed circuit. These charged particles can be either negatively charged in the form of electrons, or positively charged in the form of holes, or both. If an element contains equal amounts of both types of charge it is said to be electrically neutral.

Thus the greater the number of free electrons that move along a conductor from one point to another, the greater will be the transfer of electric charge. The good thing about electric charge is that it can be stored using batteries, capacitors, etc. for use at a later date.

However, the amount of electric charge an element possesses is very small with the smallest amount of charge being that of a single electron. So instead the unit of electric charge is called the Coulomb, (C), after Charles Coulomb.

The symbol given to represent the quantity of electric charge is the uppercase “Q”, and a lowercase “q” for charge which varies with time. The charge of just one electron is given as being: 1.6×10^{-19} coulombs, so therefore one Coulomb of charge is equal to $1/1.6 \times 10^{-19}$ or **6.25×10^{18}** electrons.

$$Q = 1 \text{ coulomb} = 6.25 \times 10^{18} \text{ electrons}$$

The flow of electric charge, Q around a closed circuit over time is called an electric current. As current is measured by the number or rate at which free electrons pass a particular point within a circuit per second, electric current can also be defined as the unit of charge per unit of time in second.

Thus when electric charge moves at the rate of 6.25×10^{18} electrons flowing past a given point per second, the value of the current past a point is one ampere. This is the same as saying one coulomb of charge per second.

The SI (International System of Units) unit of current is the ampere with letter symbol “A”. Thus a constant current has uppercase symbol **I**, while a time-varying current has the lowercase symbol **i** for intensity.

Mathematically we can define the relationship between charge (Q) and electric current (I) as being:

$$I(\text{amperes}) = \frac{Q(\text{coulombs})}{t(\text{seconds})}$$

Where:
I = Average Current flowing
Q = Total Charge passing a Fixed Point
t = Time taken to Pass a Fixed Point

Thus the relationship of $I = Q/t$ shows that one ampere equals a rate of flow of charge equal to 1 coulomb per second, and 2 amperes of current exists when 2 coulombs of electric charge pass the same point in 1 second.

In solid metals only negatively charged free electrons move to produce a current flow, the positive protons cannot move. But in a liquid or a gas, both the positive protons and negative electrons can move.

Although electric charge can be continuously transferred between different parts of a circuit, the total amount of charge (electrons or protons) does not change as charge is neither created nor destroyed.

An electric current of one ampere flows in a circuit when a charge of one coulomb passes a given point in a circuit in one second

Electric current is a measure of how concentrated or intense the flow of electrons past a point are. Current is expressed in units of amperes (A) or kiloamperes ($1\text{kA} = 10^3 \text{ A}$) for large values of current and sub-multiples of: milliamperes ($1\text{mA} = 10^{-3} \text{ A}$), microamperes ($1\mu\text{A} = 10^{-6} \text{ A}$), nanoamperes ($1 \text{ nA} = 10^{-9} \text{ A}$), or picoamperes ($1 \text{ pA} = 10^{-12} \text{ A}$).

Electric current, (A) always has a definite direction of flow associated with it. On circuit diagrams and electronic components, there is usually an arrow associated with it to indicate the direction of current flowing through or around it. Generally, the arrow used shows the direction of positive current flow, which is not necessarily the actual direction of current flow.

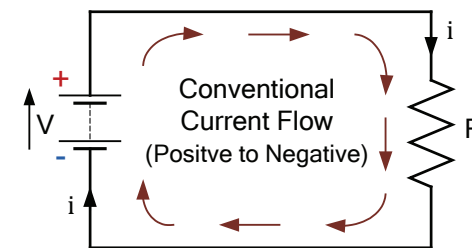
Electric current will not flow through a conductor unless there is a “closed path” for it to follow from the voltage source and back again. But there also needs to be a potential difference or p.d. across the voltage source to assist in the movement or flow of the charge.

6. CONVENTIONAL CURRENT FLOW

Conventional current flow is the “convention” from way back in the early days of electrical engineering that electrical current was the result of a positive charge moving around a closed circuit from the positive (+ve) terminal of the voltage source and back to the negative (-ve) terminal of the voltage supply. That is from positive to negative.

For DC circuits, the voltage source is generally a battery that provides a sufficient potential difference between two points to allow current in the form of electric charge to flow. Thus for a battery supply, the flow of electric charge are electrons rather than holes.

Figure 4. Conventional Current Flow



Conventional current flow, or just simply known as “current flow”, as shown in Figure 4, flows from a point of higher potential to a point of lower potential.

Thus the movement of the positive charge (holes) around a closed circuit would be from the positive terminal of the battery, through the circuit and load returning back to the negative terminal of the battery.

Then we can think of conventional current flow as positive charges in motion and we analyse electrical circuits in terms of conventional current flow because positive terminals and voltages are considered first before a negative terminal or ground.

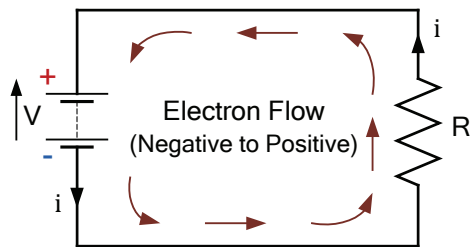
To continue with this line of thought, in all circuit diagrams and schematics, the arrows shown on symbols for components such as diodes, transistors and current sources always point in the direction of conventional current flow.

7. ELECTRON FLOW

While we may think of the flow of current around a circuit as conventional current flow from positive to negative, in a DC circuit in which a battery or rectifier provides the voltage source, the flow of electric charge around the circuit are actually electrons, which themselves have a negative charge.

Electrons move or flow around a circuit from a point that has an excess of electrons to a point that has a deficiency of electrons. Thus electrons actually flow from a negative terminal towards a positive terminal (point, or electrode).

FIGURE 5. ELECTRON FLOW



Thus the flow of electrons around the circuit is opposite to the direction of the conventional current flow being negative to positive.

This is because the actual current flowing in an electrical circuit is composed of electrons which flow from the negative terminal of the battery

source, and return back to the positive terminal of the same battery.

As the charge on an electron is negative by definition, it will always be more attracted to a positive terminal. Therefore, the flow of electrons around a closed circuit is called Electron Flow because the physical flow of electrons from negative to positive as shown in Figure 5.

We have seen that conventional current flow is in the direction in which positive charge moves from a positive (+ve) point to a more negative (-ve) point in a circuit. While electron flow is in the opposite direction from a more negative to a more positive point. The arrow in a circuit always specifies the direction of positive current flow.

Both conventional current flow and electron flow are used by many textbooks. In fact, it makes no difference which way the current is flowing around the circuit as long as the direction is used consistently throughout.

Conventional current flow is from Positive to Negative. Electron flow is from the Negative to Positive terminal

The direction of current flow does not affect what the current does within the circuit, but generally it is much easier to understand what is happening using conventional current flow – positive to negative.

8. POTENTIAL DIFFERENCE OR VOLTAGE

Voltage, (**V**) is the potential energy of an electrical supply stored in the form of an electrical charge. Voltage can be thought of as the driving force behind an electrical circuit that pushes electrons through a conductor. So the greater the voltage the greater is its ability to “push” the electrons around a given circuit.

When two positive charges or two negative charges are brought near to each other they will repel while a positive and negative charge are attracted to each other as opposites attract.

Then a charged particle has the ability to do work and this potential energy can be described as the work required in joules to move electrons in the form of an electrical current around a circuit from one point or node to another. In other words, energy must be expended, that is, WORK must be done in order to move an electric charge from one point to another.

The ability of a charged particle to do work is called an electric potential. Thus two dissimilar charges have a difference of potential and the SI unit of potential difference (p.d.) is called the volt (V), which is equal to joules per coulomb.

So if electric charge is measured in coulombs (C) where from above 1 C is equivalent to the charge on 6.24×10^{18} electrons, then they need 1 joule of input energy to create a potential difference of 1 volt.

The potential difference between two points will be one volt if one joule of work is done displacing one coulomb of charge

In electrical engineering, the potential difference across two points, connections or junctions within a circuit is commonly referred to as voltage, given the symbol **V**, (named after Alessandro Volta).

Thus the standard symbol “V” represents any potential difference between two fixed points with one of those points being a reference point for all other measurements. Whether it is taken from a source to ground (zero volts), or across a component or device.

The voltage difference between two points is the work done in joules (J) required to move one coulomb (C) of charge from one point in a circuit to another through a potential difference of one volt (1V). The SI unit of voltage is the volt and is given as:

$$V(\text{volts}) = \frac{W(\text{joules})}{Q(\text{coulombs})}$$

Where:

V = The Voltage in Volts

W = The Work being Done

Q = The Total Charge passing a Fixed Point

The unit of potential difference between two points is the Volt (V) with voltages usually expressed in volts (V), kilovolts ($1 \text{ kV} = 10^3 \text{ V}$) or megavolts ($1 \text{ MV} = 10^6 \text{ V}$) for larger voltage values, and smaller sub-multiples of: millivolts ($1 \text{ mV} = 10^{-3} \text{ V}$), or microvolts ($1 \text{ uV} = 10^{-6} \text{ V}$) for very small values of voltage.

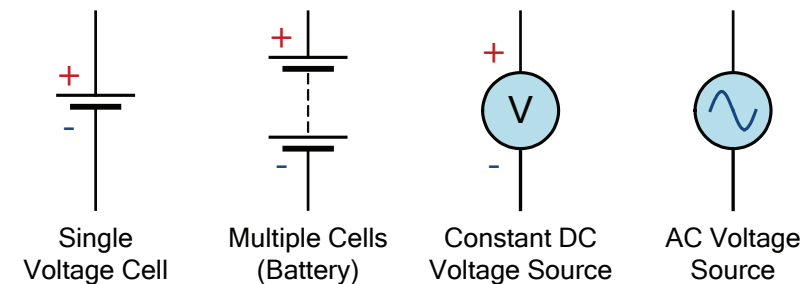
A constant voltage source is called a DC Voltage whereas a voltage that varies periodically over time being called an AC voltage. Batteries or power supplies are mostly used to produce a steady D.C. (direct current) voltage source such as 5v, 12v, 24v etc in electronic circuits and systems.

In AC (alternating current) voltage sources are available for domestic house and industrial power and lighting as well as power transmission. The mains voltage supply in the United Kingdom is currently 230 volts AC and 110 volts AC in the USA. Thus voltage can be either positive or negative.

There are two distinct ways of indicating the voltage difference between two points or nodes in a circuit. One way is presented as a straight line with an arrow pointing towards the terminal whose voltage is at a higher potential than that of the other terminal, and the other way is use of a plus (+) symbol to indicate the higher voltage point and a negative (-) symbol to indicate the lower voltage point, such as those found on a battery.

Commonly, electronic circuits operate on low voltage DC battery supplies of between 1.5V and 24V dc The circuit symbol for a constant voltage source usually given as a battery symbol with a positive, (+) and negative, (-) sign to indicate the direction of the polarity. The circuit symbol for an alternating voltage source is usually a circle with a sine wave inside as shown in figure 6.

FIGURE 6. VOLTAGE SOURCE SYMBOLS



While voltage is measured in volts, with one volt can be defined as the electrical pressure required to force an electrical current of one ampere through a resistance of one Ohm. Voltage can be either positive or negative.

$$V \text{ (volts)} = I \text{ (current)} \times R \text{ (resistance)}$$

Note that voltage can exist across a circuit without current, but current cannot exist without voltage and as such any voltage source whether DC or AC likes an open or semi-open circuit condition but hates any short circuit condition as this can destroy it.

9. CELLS AND BATTERIES

Batteries are everywhere. They come in all shapes and sizes ranging from small single button cells to large high voltage batteries to power our phones, cars, laptops and electronic circuits. A battery is an electrochemical device full of different chemical elements, conductive cathodes and insulating electrolytes which uses chemical reactions within itself to produce a flow of electrons to power a connected load.

FIGURE 7. A BUTTON OR COIN CELL



The term Battery actually means that there are two or more individual voltage cells electrically connected together within a single package to produce a single device with more voltage, current and therefore power. That is, voltage cells are the building blocks of batteries, and the more cells connected together, the greater the power available.

Then a battery cell is a single encased electrochemical device consisting of a positive (+ve) terminal called the anode, and a negative (-ve) terminal called the cathode producing a

potential differential (voltage) across its two terminals.

Individual battery cells are generally rated at 1.2 VDC or 1.5 VDC per cell voltage. However, some newer Lithium ion (Li-Ion) type cells are rated at 3 VDC per cell.

The batteries which have higher voltage output, for example 6V, 9V, 12V, etc. are constructed using many 1.5V cells in series and then encased in a standard packaging case. Batteries which have two or more voltage cells are commonly referred to as battery packs, battery modules or battery assemblies which have the primary function of providing a source of electric energy to power a given circuit or equipment.

As well as a fixed DC voltage, batteries also have a rated amperage capacity given in Milliampere-hours (mAh) or Ampere-hours (Ah) by which the batteries cells can supply a maximum amount of electric current over a time period of 1 hour until the batteries voltage cut-off point is reached.

A Battery is a DC voltage source containing one or more individual voltaic cells

Thus a 1Ah rated battery can supply 1 ampere of electric current for 1 full hour, or one-half an ampere of electric current for 2 hours, (that is half the current for twice as long).

Electrochemical batteries can be divided into two main types: Primary batteries and Secondary batteries. Primary batteries are your typical and common remote control, flashlight, toy or clock battery which are used once and replaced when exhausted (dead).

This is because the internal chemical reactions that take place to supply the electric current via the terminals when connected are irreversible, and once the stored energy is used up, there is no more.

Secondary batteries on the other hand such as car batteries, phone batteries, laptop batteries, etc. can be recharged and reused when exhausted. This because they use reversible chemical reactions inside to recharge, or restore their energy storage by reversing the flow of electrical current into them using a battery charger as a constant current source.

Secondary batteries are commonly referred to as rechargeable batteries or energy storage batteries. Secondary batteries are capable of many deep recharging cycles and high power density for use in many electrical applications.

9.1 Connecting Batteries in Series

Many of the common everyday batteries such as those referred to as: AA, AAA, C and D cells, are 1.5 volt rated batteries. However, in many electronic circuits and projects we require slightly higher voltages values. For example, 9 volts or 12 volts, etc.

Fortunately just like cells, complete batteries can also be wired together in series or parallel configurations to increase the voltage, current, or both, for a given combination.

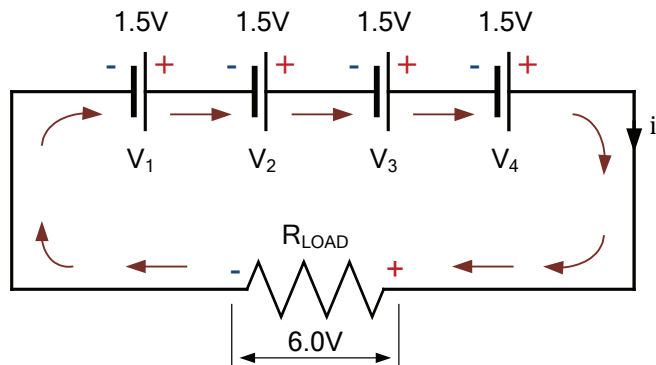
Two or more batteries in series add their voltages together, that is:

$$V_T = V_1 + V_2 + V_3 + V_4 + \dots \text{etc.}$$

while the series current capacity stays the same.

The positive terminal of one battery is connected to the negative terminal of the next, and so on. Connecting batteries together in a series combination means a higher voltage for the same current as shown in figure 8.

FIGURE 8. BATTERIES IN SERIES



Since power (P) is calculated by voltage (V) x current (I) ($P = V \times I$), increasing the series voltage increases the power output for the same current. However, for batteries connected in series, each battery must have the same amp-hour (Ah) rating but can have

different terminal voltages. This is because the battery with the lowest rating will become depleted first. Therefore, series batteries must all have the same amp-hour rating.

9.2 Connecting Batteries in Parallel

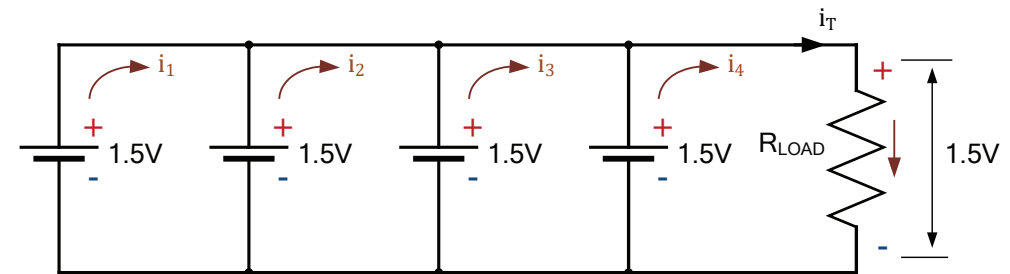
Two or more batteries that are connected in parallel produce the same output voltage as the individual batteries, but the current capacity is multiplied by the number of parallel batteries. That is parallel connected batteries will produce a greater amp-hour capacity for the same terminal voltage.

In a parallel connection, the positive terminal of one battery is connected to the positive terminal of the next with the negative terminal connected to the negative terminal.

*Series connected batteries increase terminal voltage.
Parallel connected batteries increase available current*

Connecting batteries together in parallel means a higher available current value and therefore greater amp-hour (Ah) capacity for the same terminal voltage.

FIGURE 9. BATTERIES IN PARALLEL



Again, as power (P) is calculated by voltage times current, increasing the parallel current capacity increases the power output for the same voltage. Thus two or more batteries connected in parallel add their currents together, that is:

$$I_T = I_1 + I_2 + I_3 + I_4 + \dots \text{etc.} \text{ while the terminal voltage remains the same.}$$

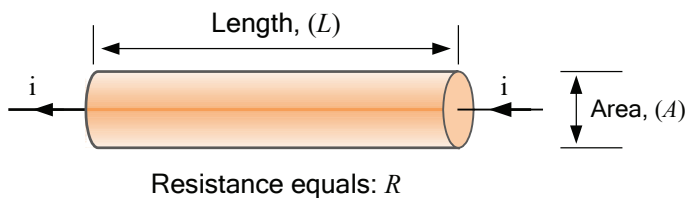
However, for batteries connected in parallel, each battery must have the same voltage rating but can have different amp-hour (Ah) current ratings. Therefore, parallel connected batteries must all have the same terminal voltage value otherwise the higher voltage battery will attempt to charge the lower voltage battery resulting in circulating currents between batteries.

10. RESISTIVITY OF A CONDUCTOR

The amount of electrical current which can flow around a closed circuit is restricted in general, by the amount of resistance (**R**) present. But the electrical resistance between two circuit points can depend on many factors such as the conductor's length, its cross-sectional area, the temperature, as well as the actual material from which it is made.

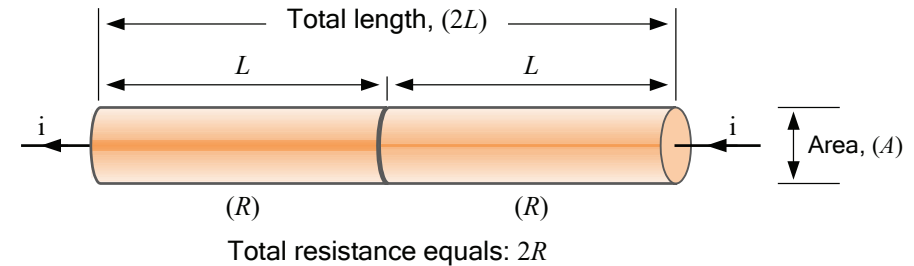
The resistivity of a particular material is the resistance it offers to the flow of an electric current through it with some materials resisting the current flow more than others. Let's firstly assume we have a piece of wire (a conductor) that has a length L , a cross-sectional area A and a resistance R as shown in Figure 10.

FIGURE 10. A SINGLE CONDUCTOR



Thus the electrical resistance (R), of a single wire conductor will be a function of its length (L), and its cross-sectional area (A) as shown. Now suppose we connect two identical conductors of Figure 10 together in series, that is end-to-end as shown in Figure 11.

FIGURE 11. DOUBLING THE LENGTH OF THE CONDUCTOR

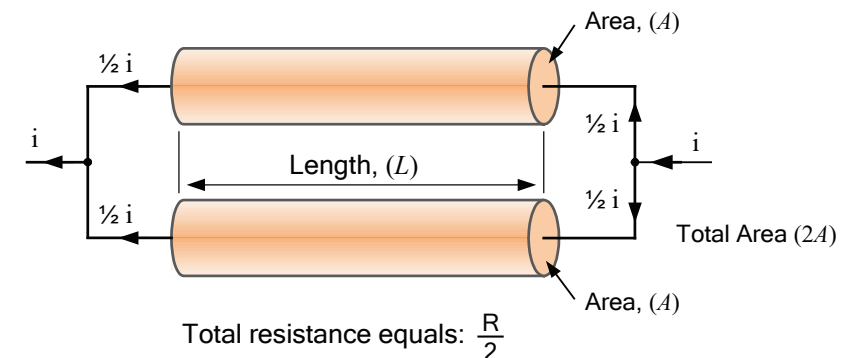


With the two single conductors connected together in a series, that is end to end, we have effectively doubled the overall length of the conductor ($2L$), while the cross-sectional area (A), remains exactly the same as before.

But as well as doubling the overall length, we have also doubled the total resistance of the conductor to current flow, giving $2R$ as: $1R + 1R = 2R$. Therefore, we can see that the resistance of the conductor is proportional to its length, that is: $R \propto L$. In other words, we would expect the electrical resistance of a conductor to be proportionally greater the longer it is.

Now suppose we connect two identical single conductors from above together in parallel as shown in figure 12.

FIGURE 12. DOUBLING THE AREA OF THE CONDUCTOR



By connecting the two conductors together in a parallel, we have effectively doubled the total area giving $2A$, while the conductors length, L remains the same as for the original single conductor. But as well as doubling the area, by connecting the two conductors together in parallel we have effectively halved the total resistance of the conductor, giving $1/2R$ as now each half of the current flows through each conductor branch.

Thus the resistance of the conductor is inversely proportional to its area, that is: $R \propto 1/A$. In other words, we would expect the electrical resistance of a conductor to be proportionally less the greater is the cross-sectional area of the conductor. Therefore, we can see that the resistance of a single conductor is directly proportional to its length, as: $R \propto L$, and inversely proportional to its area, as: **$R \propto 1/A$** .

But as well as length (L), and conductor area (A), we would also expect the electrical resistance of the conductor to depend upon the actual material from which it is made, because different metal elements, copper, silver, aluminium, etc. all exhibit different physical and electrical properties to the flow of electric current.

Then the resistance of a particular material, wire or conductor is commonly given as:

$$R = \rho \left[\frac{L}{A} \right] (\Omega) \text{ Ohms}$$

Where: R is the resistance in ohms (Ω), L is the length in metres (m), A is the area in square metres (m^2), and the proportional constant ρ (the Greek letter “rho”) is known as Resistivity.

Note that resistivity (ρ), is always a constant value of the type of material being calculated, with the resistivity of the material given in ohm-meter ($\Omega\cdot m$). For example, the resistivity of copper at room temperature ($20^\circ C$) is generally given as being: 1.7×10^{-8} ohm-meters.

So depending upon the electrical resistivity value of a particular material, it can be classified as being either a “conductor”, an “insulator” or a “semiconductor”. Remember from above that semiconductors are materials where its conductivity is dependent upon the impurities added to the material.

Also, the resistivity of a particular copper cable is important to know when calculating the I^2R voltage drop of a conductor. This allows the correct wire cross-sectional size to be chosen, especially in solar installations using DC supplies.

11. ELECTRICAL RESISTANCE

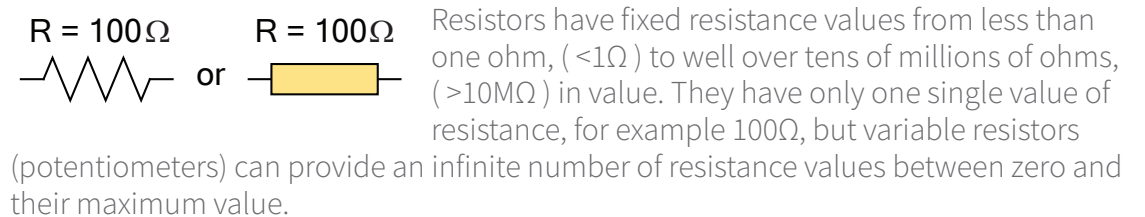
The current flowing in an electrical circuit not only depends upon the voltage pushing the current around, but also on the resistance of the wires, connections and components that make up the electrical circuit. Resistance is the ability of a material to resist or regulate the flow of current within a circuit. Thus resistance opposes current flow.

The electrical circuit element which does this perfectly is called a resistor. The resistor is a fundamental component of a circuit, and the amount of “resistance” a resistor has is measured in units called Ohms (Ω) with its value denoted by uppercase letter “ R ”, for resistance.

Electrical resistance is the opposition to current flow around a closed circuit

In all Electrical and Electronic circuit diagrams and schematics, the most commonly used symbol for a fixed value resistor is that of a “zig-zag” type line with the value of its resistance given in Ohms, Ω as shown in Figure 13.

FIGURE 13. RESISTOR SCHEMATIC SYMBOL



The amount of resistance a resistive component has is determined by the relationship of the current through it to the voltage across it. A resistor is said to have a resistance of one ohm when one volt causes one ampere of current to flow through it. That is $R = V/I$ (this is known as Ohm’s law).

A low value resistance, for example one ohm or less implies a good conductor made from materials with lots of free electrons in its valence shell such as copper, aluminium or carbon allowing electric current to easily flow through it.

A high value resistance in the millions of ohms or more implies a bad conductor made from insulating materials such as glass, porcelain or plastics that blocks or prevents current flow.

The SI unit of resistance is the Ohm with Greek symbol Ω (Omega) with the most common prefixes used being the Kilo-ohm ($k\Omega = 10^3\Omega$) and Mega-ohm ($M\Omega = 10^6\Omega$).

PREFIXES USED FOR OHMS

Prefix for Ohms	Symbol	Description	Abbreviation	Value
Megaohm	M	One Million Ohms	$M\Omega$	$10^6 \Omega$
Kilohm	k	One Thousand Ohms	$k\Omega$	$10^3 \Omega$
Ohm	-	One Ohm	Ω	$10^0 \Omega$
Milliohm	m	One Thousandth of an Ohm	$m\Omega$	$10^{-3} \Omega$
Microhm	μ	One Millionth of an Ohm	$\mu\Omega$	$10^{-6} \Omega$

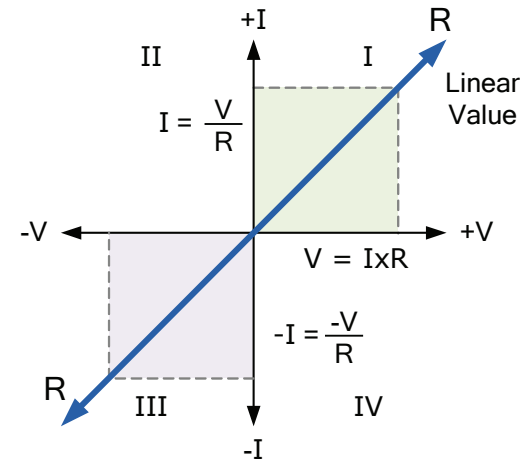
Generally, the prefix symbols used are capital letters for multipliers equal to or greater than one million (10^6), for example $1M\Omega$. Lowercase letters are used for multipliers equal too or smaller than one thousand (10^3), for example $1k\Omega$.

Resistance is always regarded as being pure and unaffected by changes in frequency. In AC circuits, AC resistance is equal in value to its DC resistance, $Z_R = R$. Also, resistance may have linear or non-linear values, but not negative, only positive.

Temperature also has an effect on the resistance of an electrical conductor. In most metallic conductors such as copper, or aluminium, their resistance increases with temperature.

Linear resistance obeys Ohm's Law as the voltage across the resistor is linearly proportional to the current through it. Non-linear resistances such as lamp filaments do not obey Ohm's Law but have a voltage drop across them that is proportional to some power of the current.

FIGURE 14. RESISTANCE I-V CHARACTERISTICS



The relationship between voltage, (V) and current, (I) in a linear circuit of constant resistance, (R) would produce a straight line I-V characteristics curve with its slope equal to the value of the resistance as shown in Figure 14.

Then we can see that electrical resistance R, is the coefficient of proportionality between the voltage and current, as I is simply proportional to V (positive or negative), thus allowing any quantity to be found whenever two of the three quantities are known.

A resistor is classed as a passive circuit element and as such cannot deliver power or store energy. Instead resistors absorbed power that appears as heat and light. Power in a resistance is always positive regardless of voltage polarity and current direction.

Resistance can also be characterised by a quantity called conductance, (G). Conductance is the ability or measure of how a conductive material can easily pass electrons through itself. That is, the ease by which current flows through a conductor or device.

12. CONDUCTANCE

Conductance is defined as being the reciprocal of the resistance, with the inverse of resistivity being called conductivity.

For very low values of resistance, for example in the milliohms, (mΩ) range, or when working with parallel connected components, it is sometimes much easier to use the reciprocal of resistance (1/R) rather than resistance (R) itself as it can be easier on the maths involved.

High values of conductance implies a good conductor such as copper while low values of conductance implies a bad conductor such as an insulator. So the less the resistance, the higher the conductance.

The standard unit of measurement given for conductance is the **Siemen**, symbol (**S**) with the unit used for conductance being mho (ohm spelt backward), which itself is symbolised by an inverted Ohm sign Ω .

$$G = \frac{1}{R} \text{ Siemens}$$

If the resistance of a conductive material is known, dividing its resistive value into 1 gives its equivalent conductance. Similarly, if the conductance is known, dividing its value into 1 gives its equivalent resistance. For example, a resistance of 100Ω would be:

$$G = \frac{1}{100} = 0.01\text{S}$$

The total conductance of any number of parallel connected resistances, is equal to the sum of all the individual conductance's. Thus total conductance is derived as:

$$G_T = G_1 + G_2 + G_3 + G_4 + \dots \text{etc.}$$

Then we can see that the current flowing through a device of conductance G with a voltage V across it is given by the equation: $I = GV$ (Ohm's law). Thus a small resistance value would give a large conductance value resulting in a large current under the influence of an applied voltage, V.

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